



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- Summary of methodologies

Data was collected from the SpaceX API and through web scraping. The collected data was then cleaned and wrangled to prepare it for analysis. Exploratory data analysis was performed using SQL and data visualization techniques. Interactive visual analytics were created using Folium, and machine learning classification models were developed to predict launch success.

- Summary of all results

The analysis identified important patterns in launch outcomes based on factors such as launch site, orbit type, payload mass, and booster version. Exploratory data analysis and visualizations helped reveal trends in successful and unsuccessful launches. Interactive maps and dashboard screenshots were used to present geographic and operational insights. Predictive analytics showed that machine learning models can be used to estimate launch success based on selected features.

Introduction

- Project background and context

This predicts whether the Falcon 9 first stage will land successfully. Since SpaceX launch costs are lower mainly because the first stage can be reused, predicting landing success can help estimate the overall cost of a launch.

- Problems you want to find answers

Key questions:

- Can we predict if the Falcon 9 first stage will land successfully?
- Which factors influence landing success?
- How can this prediction help estimate launch cost?
- How could competitors use this information when bidding against SpaceX?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data was collected from the SpaceX REST API and through web scraping from Wikipedia.
- Perform data wrangling
 - Missing values were handled, unnecessary columns were removed, and a target variable was created to show whether the Falcon 9 first stage landed successfully.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Classification models such as Logistic Regression, SVM, Decision Tree, and KNN were trained and tuned using GridSearchCV.

Data Collection

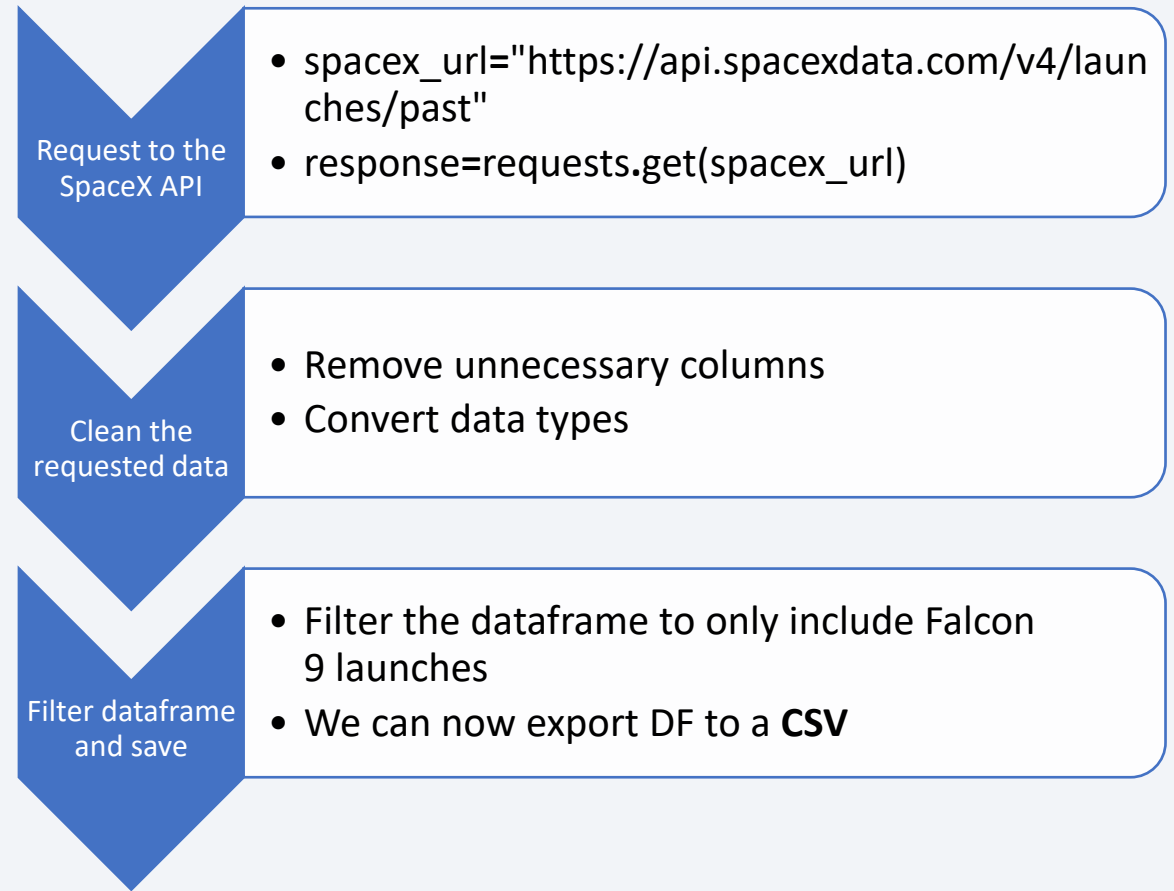
- Describe how data sets were collected.

The datasets were collected from two main sources: the SpaceX REST API and web scraping from Wikipedia. The SpaceX API provided structured launch data, including launch dates, rocket information, payload details, orbit types, launch sites, and landing outcomes. Wikipedia launch tables were scraped to collect additional historical launch records and support the API data.

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

https://github.com/laptop-user/Data_Science_Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Performed web scraping to collect Falcon 9 historical launch records from a Wikipedia page:

["https://en.wikipedia.org/wiki/List of Falcon 9 and Falcon Heavy launches"](https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)

- [https://github.com/laptop-user/Data Science Capstone/blob/main/jupyter-labs-webscraping.ipynb](https://github.com/laptop-user/Data_Science_Capstone/blob/main/jupyter-labs-webscraping.ipynb)

Request the Falcon9
Launch Wiki page

- `static_url="https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"`
- `response=requests.get(static_url,headers=headers)`

Create
a BeautifulSoup object

- `soup=BeautifulSoup(response.text,"html.parser")`

Extract table with
HTML table header

- `html_tables=soup.find_all('table')`
- `first_launch_table=html_tables[2]`
- Create a data frame and export DF to a **CSV**

Data Wrangling

Perform exploratory Data Analysis and determine Training Labels

https://github.com/laptop-user/Data Science Capstone/blob/main/Data_wrangling.ipynb

Identify missing values and Data types

- the missing values were handled
- Identify which columns are numerical and categorical

Calculate several Space X features

- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Calculate the number and occurrence of mission outcome of the orbits

Determine Training Labels

- Create a landing outcome label from Outcome column
- Create a data frame and export DF to a **CSV**

EDA with Data Visualization

- Exploratory Data Analysis
 - Scatter point chart to visualize the relationship between Flight Number and Launch Site
 - Scatter point chart to visualize the relationship between Payload Mass and Launch Site
 - Bar chart to visualize the relationship between success rate of each orbit type
 - Scatter point chart to visualize the relationship between Flight Number and Orbit type
 - Scatter point chart to visualize the relationship between Payload Mass and Orbit type
 - Line plot to visualize the launch success yearly trend
- Preparing Data Feature Engineering
 - Apply OneHotEncoder to the column Orbits, LaunchSite, LandingPad, and Serial will be used in success prediction in the future module.
- [https://github.com/laptop-user/Data_Science_Capstone/blob/main/EDA with Visualization Lab.ipynb](https://github.com/laptop-user/Data_Science_Capstone/blob/main/EDA_with_Visualization_Lab.ipynb)

EDA with SQL

- Reviewed all launch records and identified unique launch sites.
- Filtered launch records by site name, including sites beginning with “ CCA. ”
- Calculated payload statistics, including total payload for “NASA (CRS) ” and average payload for “ F9 v1.1. ”
- Identified key landing outcomes, including the first successful ground pad landing.
- Compared successful and failed mission outcomes.
- Analyzed boosters by payload mass, landing outcome, launch site, and date.
- Ranked landing outcome counts between “2010-06-04 ” and “2017-03-20”
- https://github.com/laptop-user/Data_Science_Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Added markers for all SpaceX launch sites to show their geographic locations.
- Added colored markers/circles to show successful and failed launches at each site.
- Used different colors to make launch outcomes easy to compare visually.
- Added lines and distance labels from launch sites to nearby features such as coastlines, cities, railways, and highways.
- Calculated these distances to understand how close each launch site is to important surrounding locations.
- The map helped reveal launch site locations, outcome patterns, and nearby geographic factors that may be relevant to launch operations.

https://github.com/laptop-user/Data_Science_Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Dashboard Summary

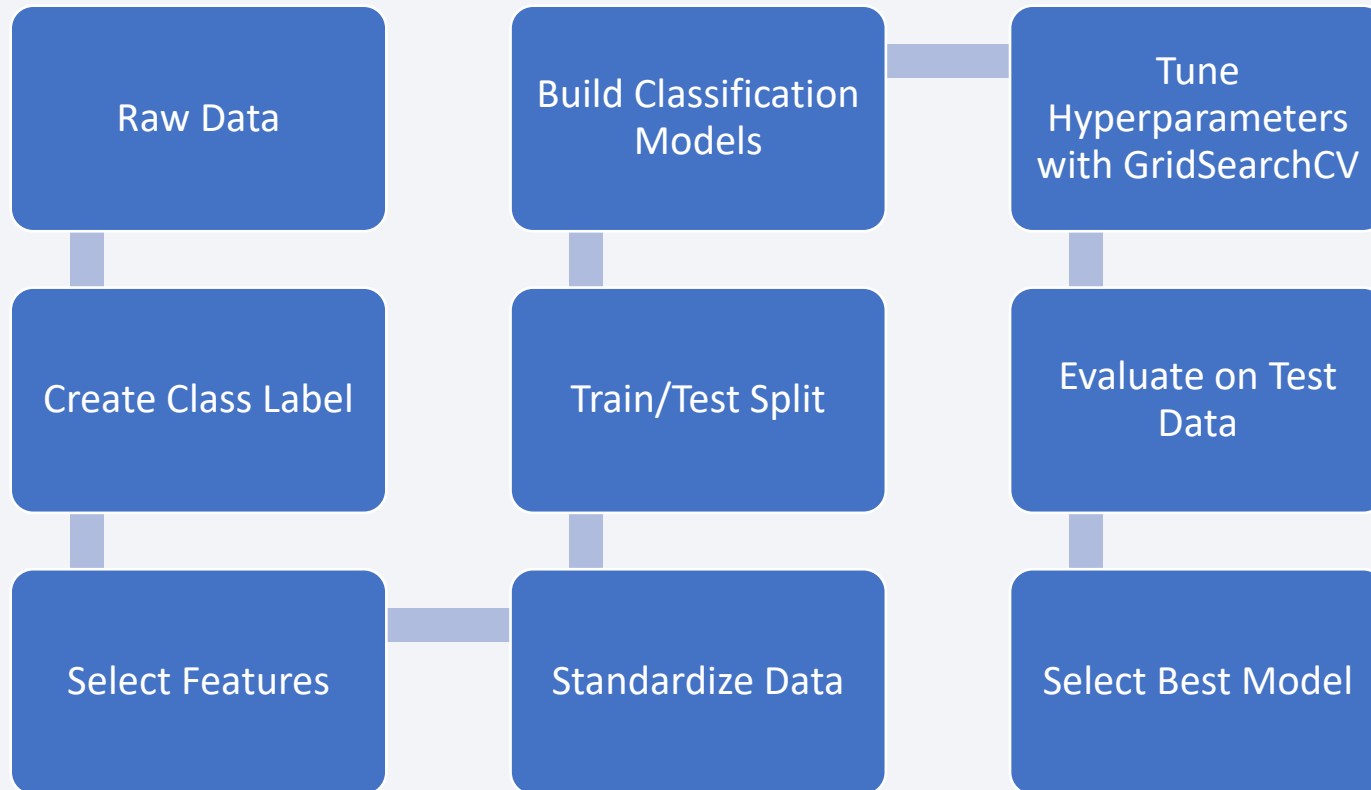
- Built an interactive Plotly Dash dashboard using SpaceX launch data.
- Added a launch site dropdown to view all sites or one selected site.
- Added a pie chart to compare successful and failed launches.
- Added a scatter plot to show the relationship between payload mass and launch success.
- Added a payload range slider to filter launches by payload mass.
- Used callbacks so charts update automatically based on user selections.

Purpose

- The dashboard helps users explore launch success by site and payload range.
- It makes patterns in launch outcomes easier to compare visually.

https://github.com/laptop-user/Data_Science_Capstone/blob/main/spacex-dash-app.py

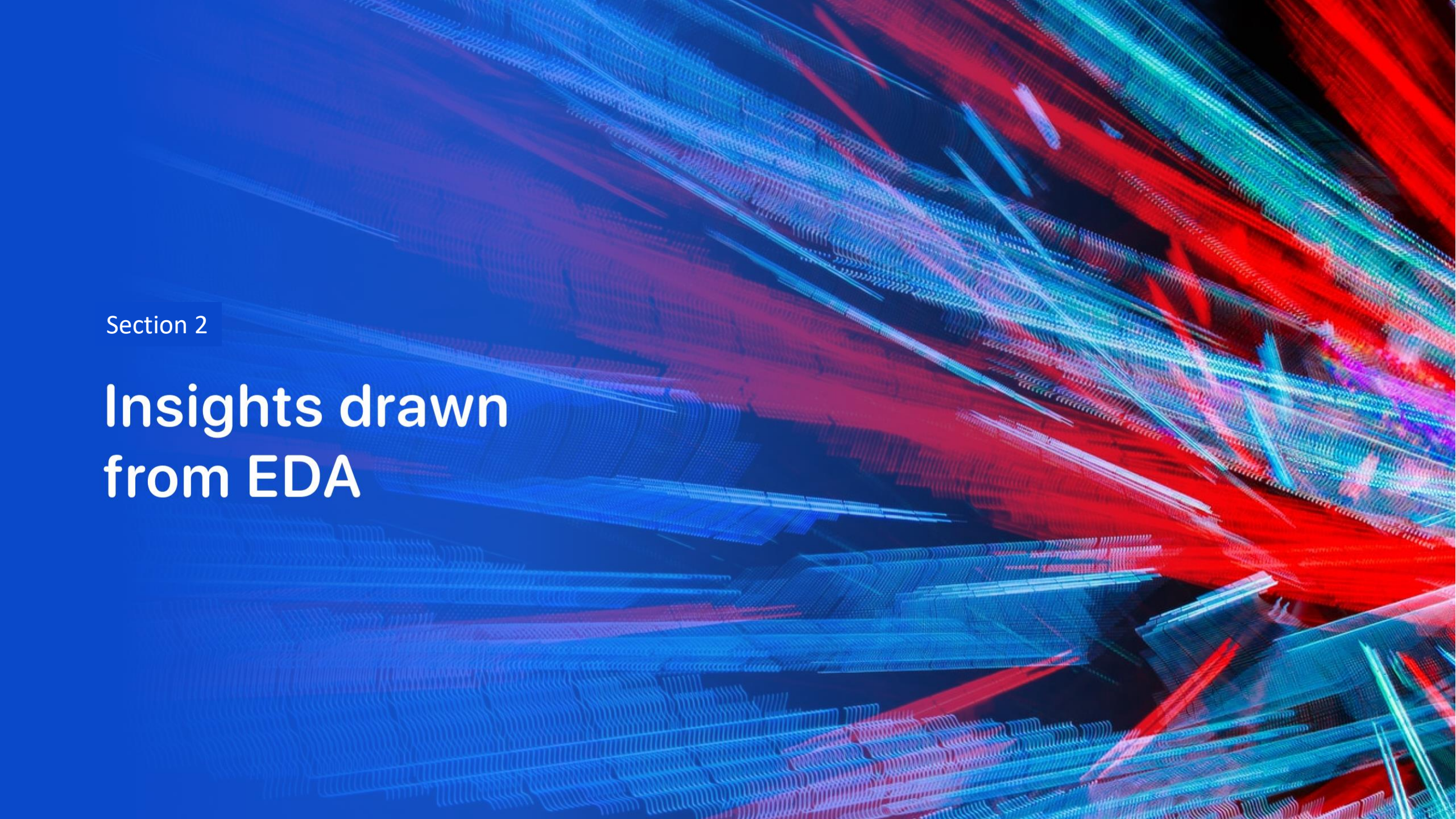
Predictive Analysis (Classification)



https://github.com/laptop-user/Data_Science_Capstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

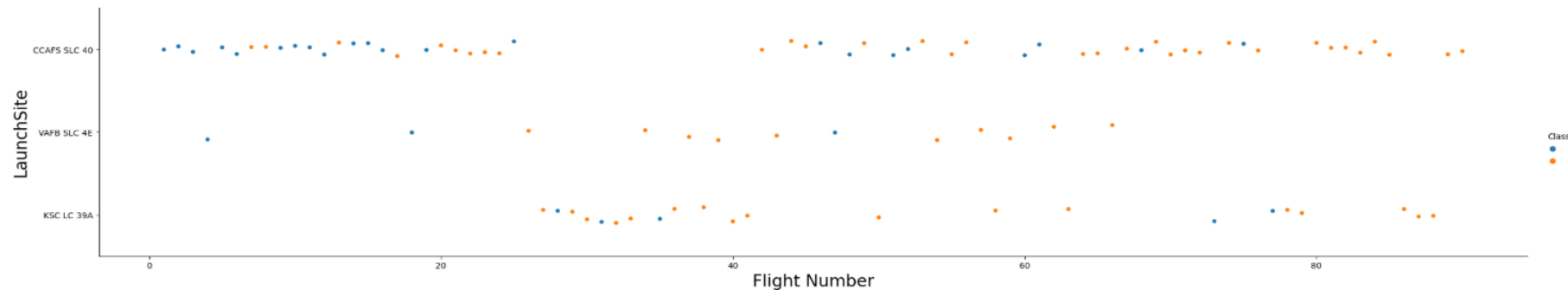
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

```
sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect = 5)  
plt.xlabel("Flight Number",fontsize=20)  
plt.ylabel("LaunchSite",fontsize=20)  
plt.show()
```

```
In [33]: sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect = 5)  
plt.xlabel("Flight Number",fontsize=20)  
plt.ylabel("LaunchSite",fontsize=20)  
plt.show()
```



- Compared flight number by launch site, using color to show successful and failed landing outcomes.
- Flights after 80 show more successes, and success rates differ by launch site.

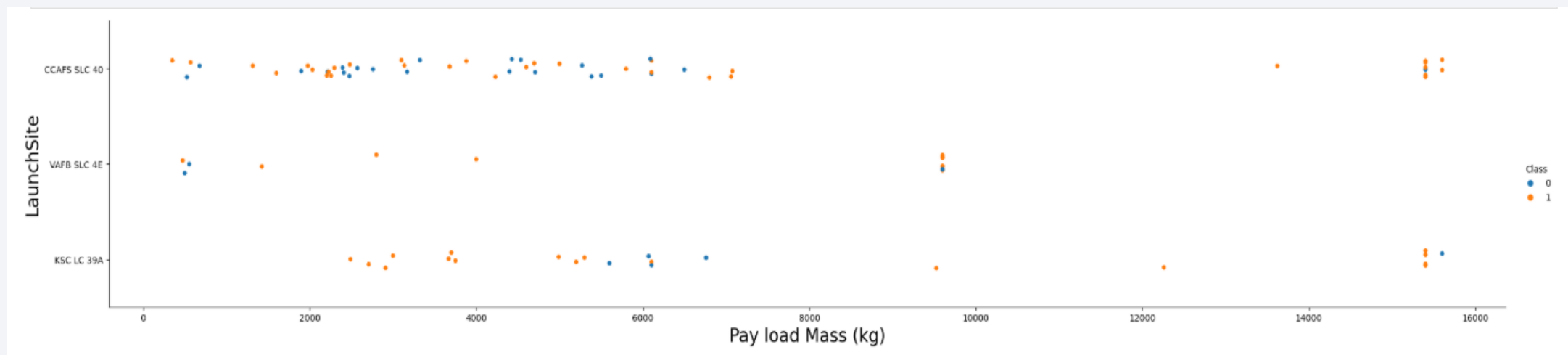
Payload vs. Launch Site

```
sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df, aspect = 5)

plt.xlabel("Pay load Mass (kg)",fontsize=20)

plt.ylabel("LaunchSite",fontsize=20)

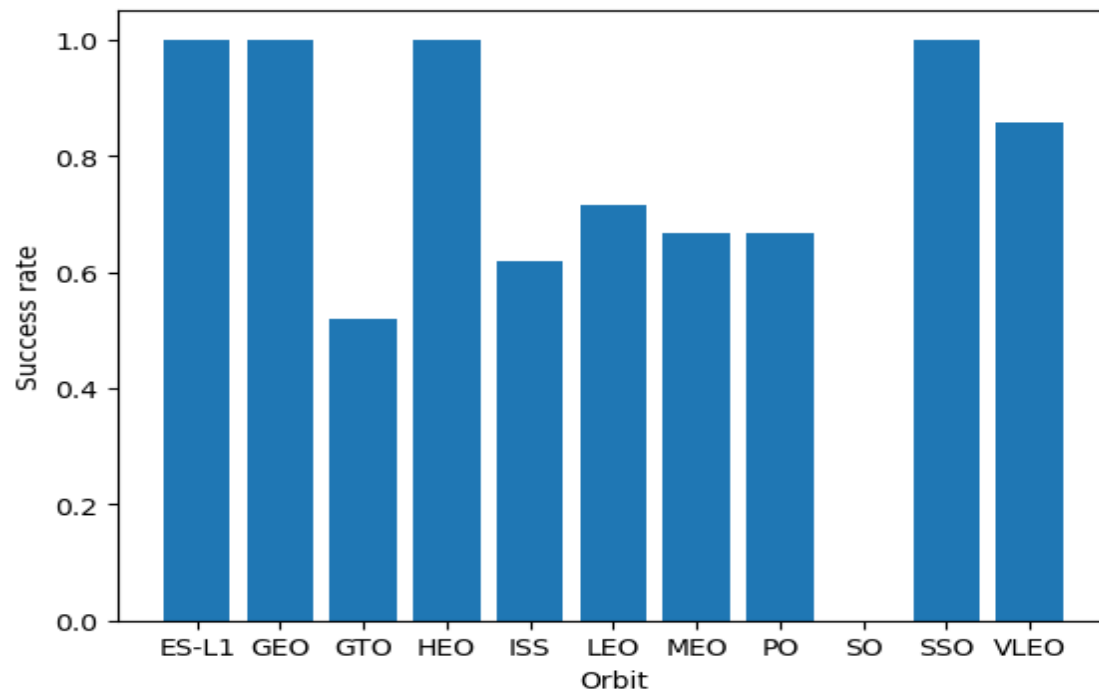
plt.show()
```



- Compared payload mass by launch site, using color to show successful and failed outcomes.
- Payloads at 'VAFB SLC 4E' are mostly under 10,000 kg, and higher payloads generally show better landing success across all sites.

Success Rate vs. Orbit Type

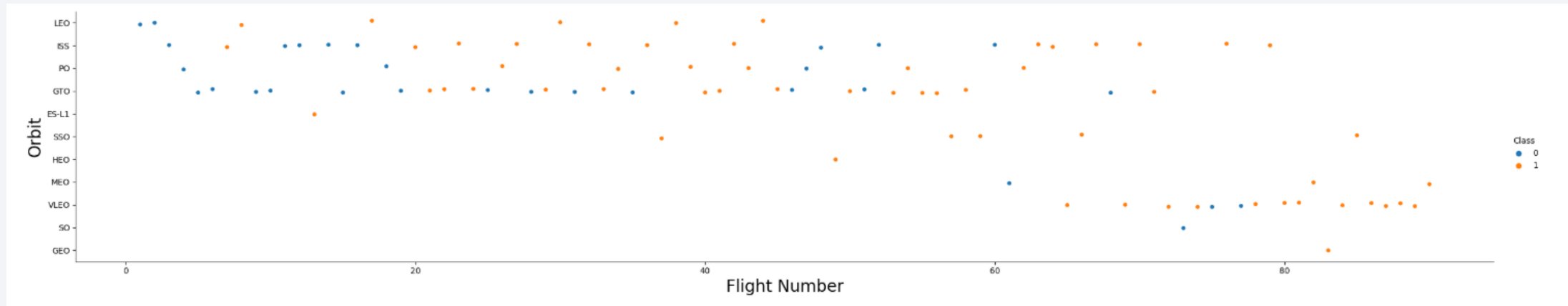
```
plt.bar(df.groupby('Orbit')['Class'].mean().keys(),df.groupby('Orbit')['Class'].mean())  
plt.ylabel('Success rate')  
plt.xlabel('Orbit')  
plt.show()
```



- ES-L1, GEO, HEO, and SSO had a 100% launch success rate.
- Most other orbits had success rates between 51% and 85%, while SO had 0% success.

Flight Number vs. Orbit Type

```
sns.catplot(y="Orbit", x="FlightNumber", hue="Class", data=df, aspect = 5)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("Orbit",fontsize=20)
plt.show()
```



- Compared flight number by orbit, using color to show successful and failed outcomes.
- Later flights show more successful landings, and newer orbits such as VLEO, SO, and GEO appear in later years.

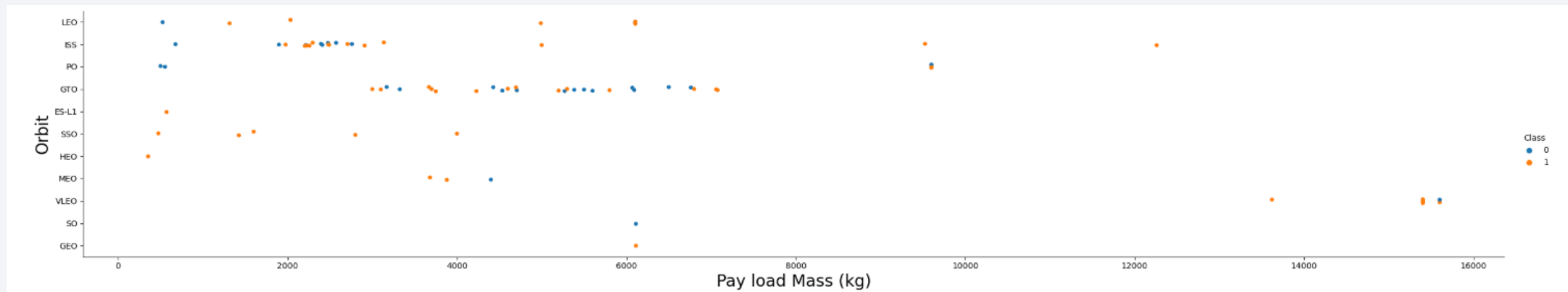
Payload vs. Orbit Type

```
sns.catplot(y="Orbit", x="PayloadMass", hue="Class", data=df, aspect = 5)
```

```
plt.xlabel("Pay load Mass (kg)",fontsize=20)
```

```
plt.ylabel("Orbit",fontsize=20)
```

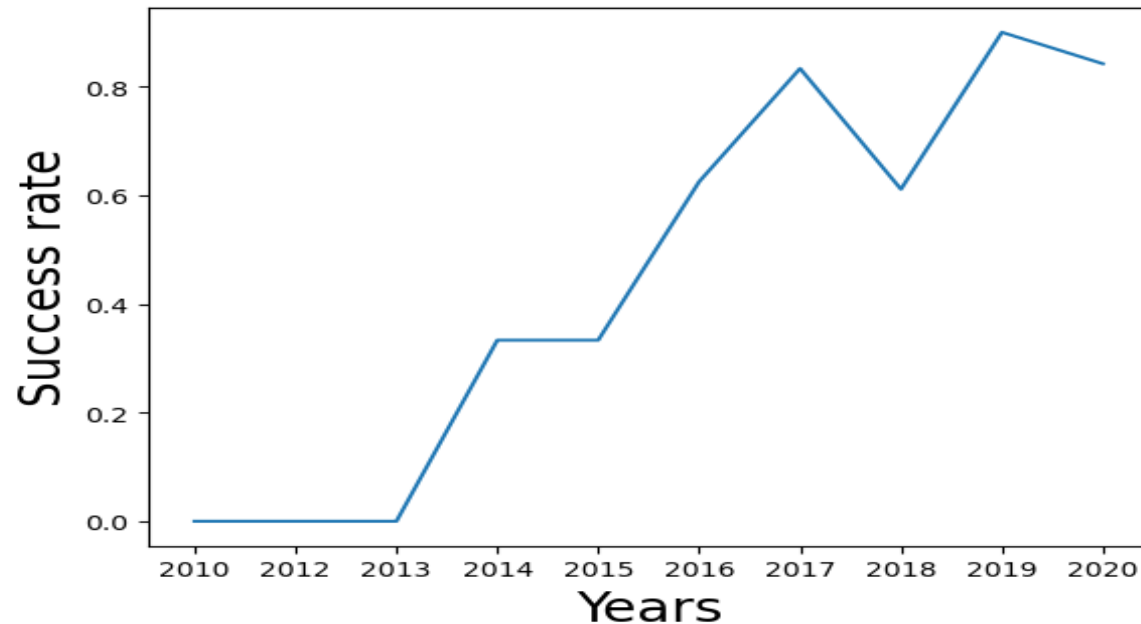
```
plt.show()
```



- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend

```
success_rate = df.groupby('Date')['Class'].mean()#.sort_values(ascending=False)
sns.lineplot(y=success_rate.values, x=success_rate.index)
plt.xlabel("Years",fontsize=20)
plt.ylabel("Success rate",fontsize=20)
plt.show()
```



The success rate since 2013 kept increasing till 2020

All Launch Site Names

This query removes duplicate launch site names and shows each launch location only once. The result helps identify where SpaceX missions were launched from.

```
[35]: %sql select DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[35]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

This query filters the dataset to show only launch sites whose names start with CCA. The LIMIT 5 clause displays only the first five matching records.

```
[33]: %sql select * FROM SPACEXTABLE where Launch_Site like 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[33]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

```
[30]: %sql select sum(PAYLOAD_MASS_KG_) FROM SPACEXTABLE;  
      * sqlite:///my_data1.db  
      Done.  
[30]: sum(PAYLOAD_MASS_KG_)  
      619967
```

The total payload mass carried by boosters launched by NASA (CRS) is 619 967 kg

Average Payload Mass by F9 v1.1

```
[34]: %sql select AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE where Booster_Version = 'F9 v1.1';
      * sqlite:///my_data1.db
      Done.
[34]: AVG(PAYLOAD_MASS_KG_)
      2928.4
```

The average payload mass carried by booster version F9 v1.1 is 2 928.4 kg

First Successful Ground Landing Date

```
[38]: %sql select min(Date) FROM SPACEXTABLE where Landing_Outcome = 'Success (ground pad)';  
* sqlite:///my_data1.db  
Done.  
[38]: min(Date)  
2015-12-22
```

- Find the dates of the first successful landing outcome on ground pad is 22 December 2015

Successful Drone Ship Landing with Payload between 4000 and 6000

```
[39]: %sql select distinct Booster_Version from SPACEXTABLE where Landing_Outcome = 'Success (drone ship)' and 4000 < PAYLOAD_MASS_KG_ < 6000;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[39]: Booster_Version
```

F9 FT B1021.1
F9 FT B1022
F9 FT B1023.1
F9 FT B1026
F9 FT B1029.1
F9 FT B1021.2
F9 FT B1029.2
F9 FT B1036.1
F9 FT B1038.1
F9 B4 B1041.1
F9 FT B1031.2
F9 B4 B1042.1
F9 B4 B1045.1
F9 B5 B1046.1

List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000:

F9 FT B1021.1, F9 FT B1022, F9 FT B1023.1, F9 FT B1026, F9 FT B1029.1, F9 FT B1021.2, F9 FT B1029.2, F9 FT B1036.1, F9 FT B1038.1, F9 B4 B1041.1, F9 FT B1031.2, F9 B4 B1042.1, F9 B4 B1045.1, F9 B5 B1046.1

Total Number of Successful and Failure Mission Outcomes

```
[44]: %sql SELECT Mission_Outcome, COUNT(*) AS count FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[44]:
```

Mission_Outcome	count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

The total number of successful mission outcomes are 100
and failure mission outcomes is 1

Boosters Carried Maximum Payload

```
[55]: %sql SELECT distinct Booster_Version, PAYLOAD_MASS_KG_ FROM SPACEXTABLE where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE)
* sqlite:///my_data1.db
Done.
```

```
[55]:
```

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

We determined the booster that have carried the maximum payload using a subquery in the WHERE clause and the MAX() function

2015 Launch Records

```
[46]: %%sql
      SELECT substr(Date, 6,2) as month, Landing_Outcome, Booster_Version, Launch_Site
      FROM SPACEXTABLE
      where substr(Date, 0,5) = '2015' and Landing_Outcome = 'Failure (drone ship)'

      * sqlite:///my_data1.db
Done.
```

```
[46]:
```

	month	Landing_Outcome	Booster_Version	Launch_Site
	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The query shows failed drone ship landing attempts in 2015. There were two failures, both launched from CCAFS LC-40 using Falcon 9 v1.1 boosters.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
[48]: %%sql
      SELECT Landing_Outcome, count(*) as count
      FROM SPACEXTABLE
      where '2010-06-04' < Date < '2017-03-20' GROUP BY Landing_Outcome ORDER BY count DESC;

* sqlite:///my_data1.db
Done.
```

```
[48]:
```

Landing_Outcome	count
Success	38
No attempt	21
Success (drone ship)	14
Success (ground pad)	9
Failure (drone ship)	5
Controlled (ocean)	5
Failure	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1
No attempt	1

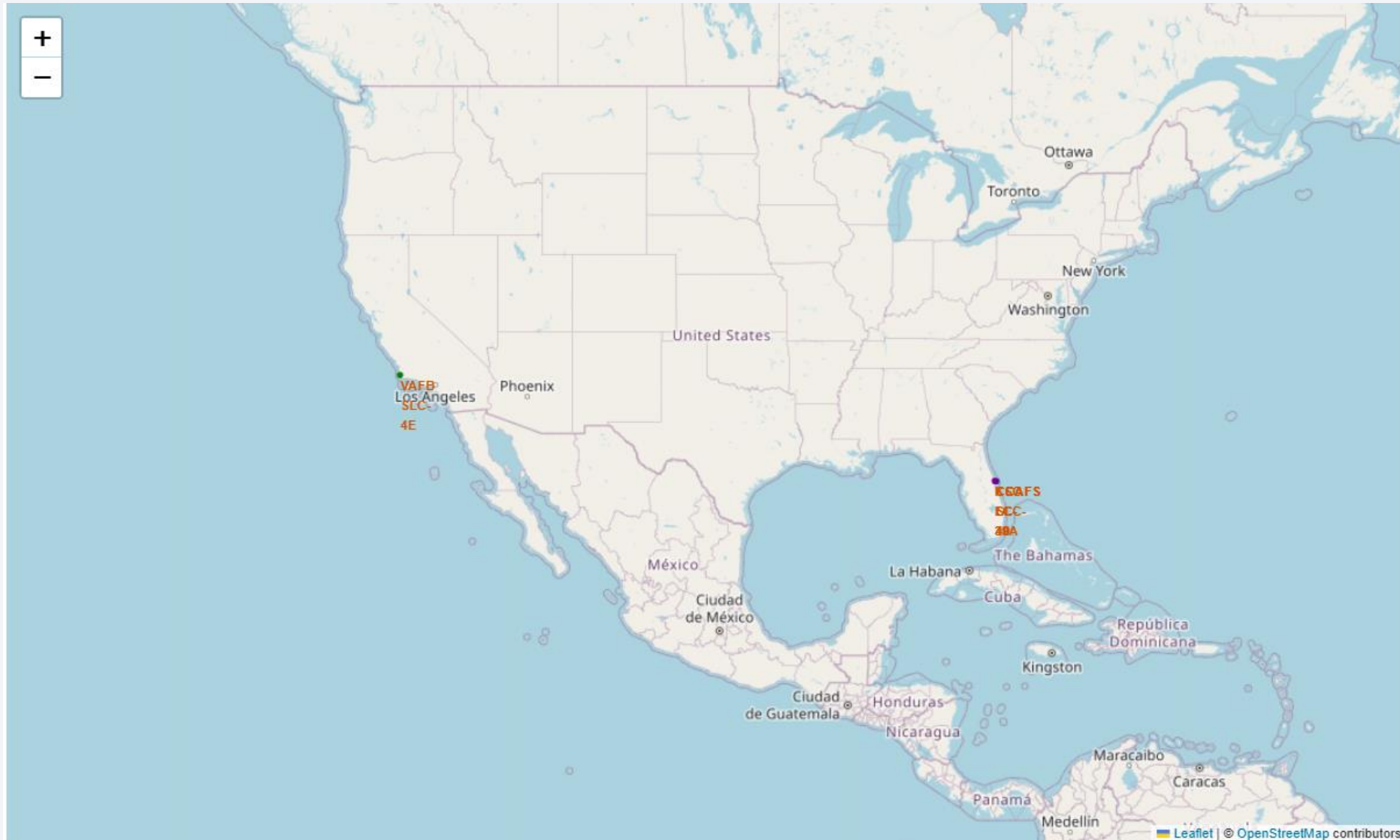
This query ranks landing outcomes by frequency during the selected date range. It shows that successful outcomes were more common than failed outcomes between 2010-06-04 and 2017-03-20.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

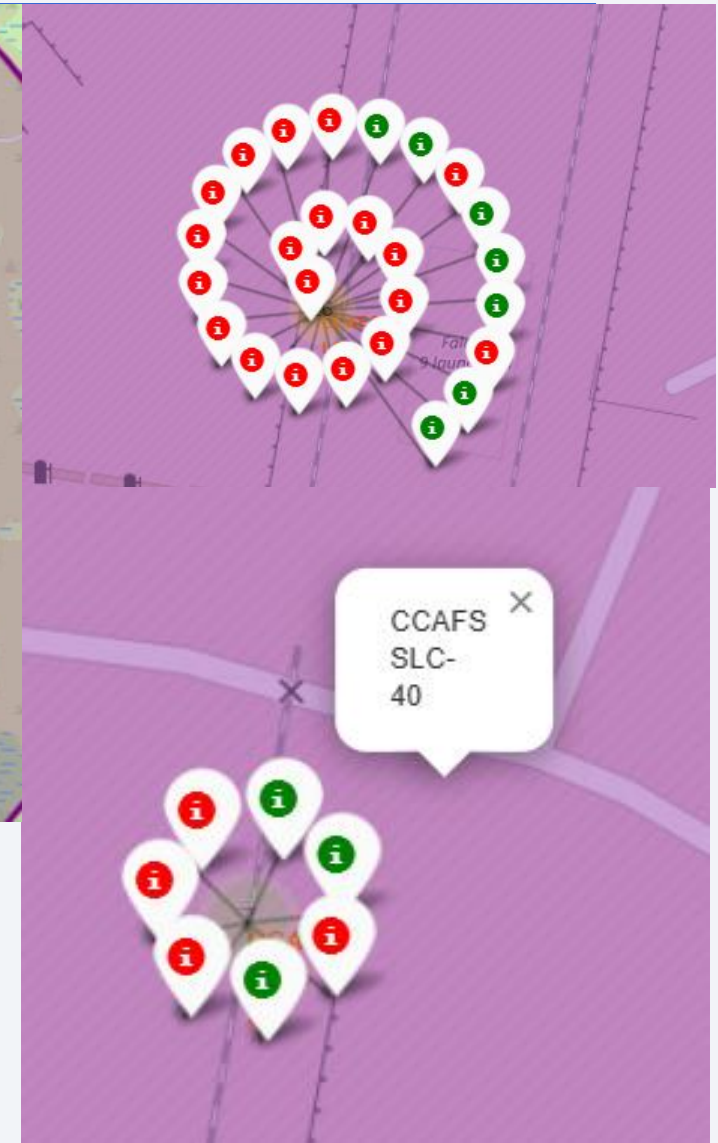
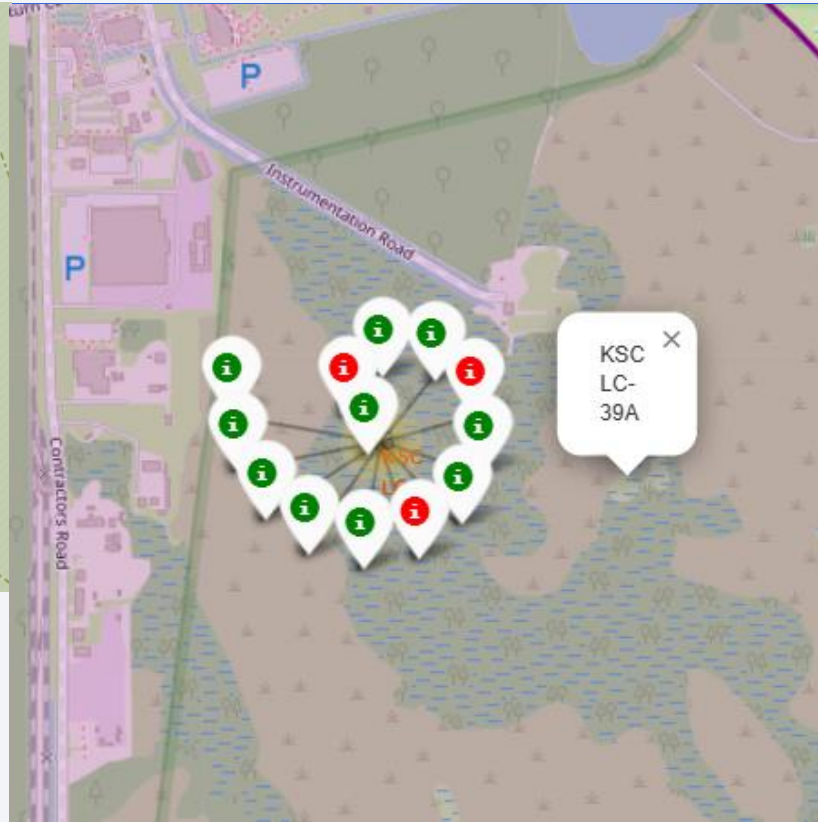
Launch Sites Proximities Analysis

All launch sites



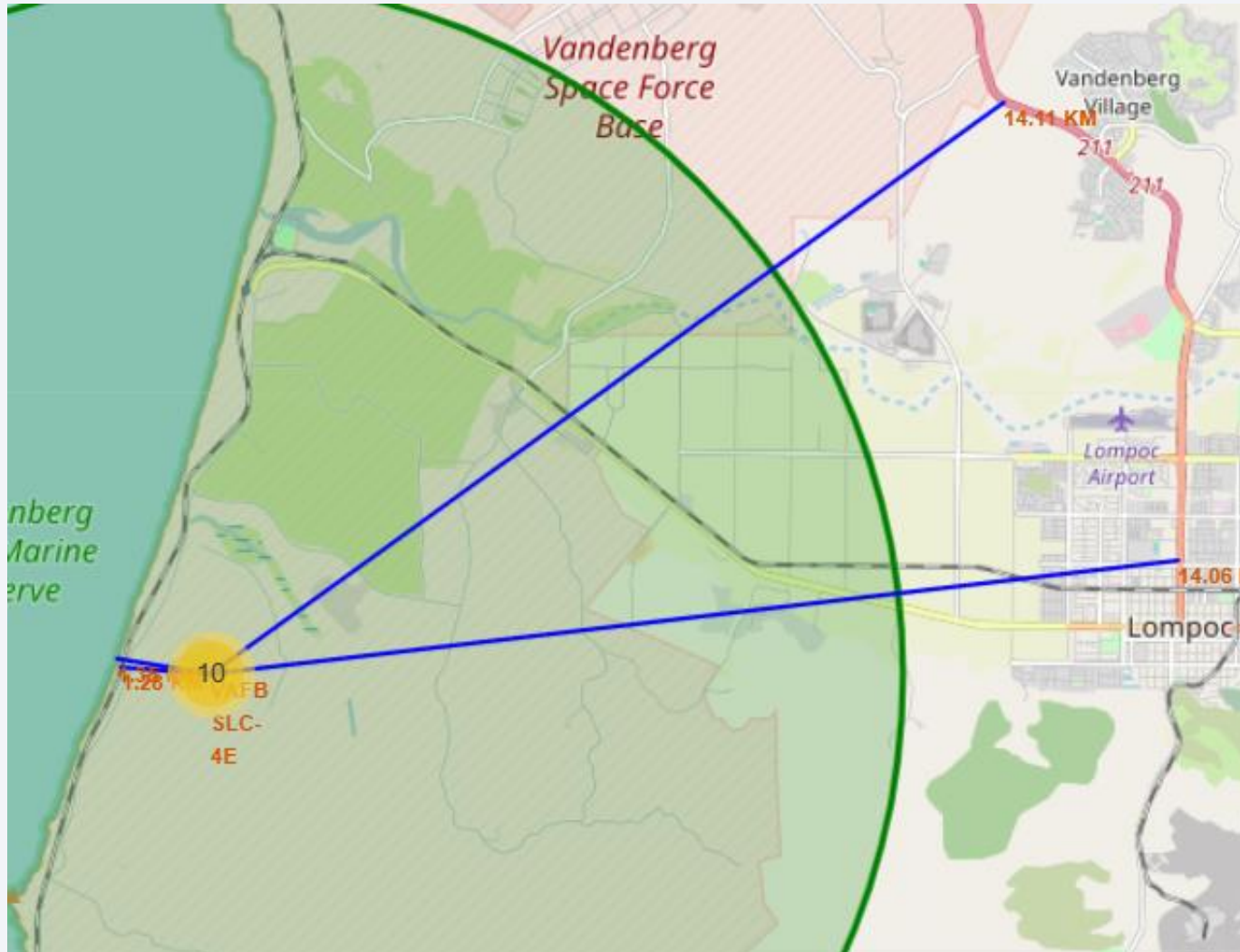
All Space X launch sites shown on the Folium map are located in the USA.

Launch Outcomes by Site

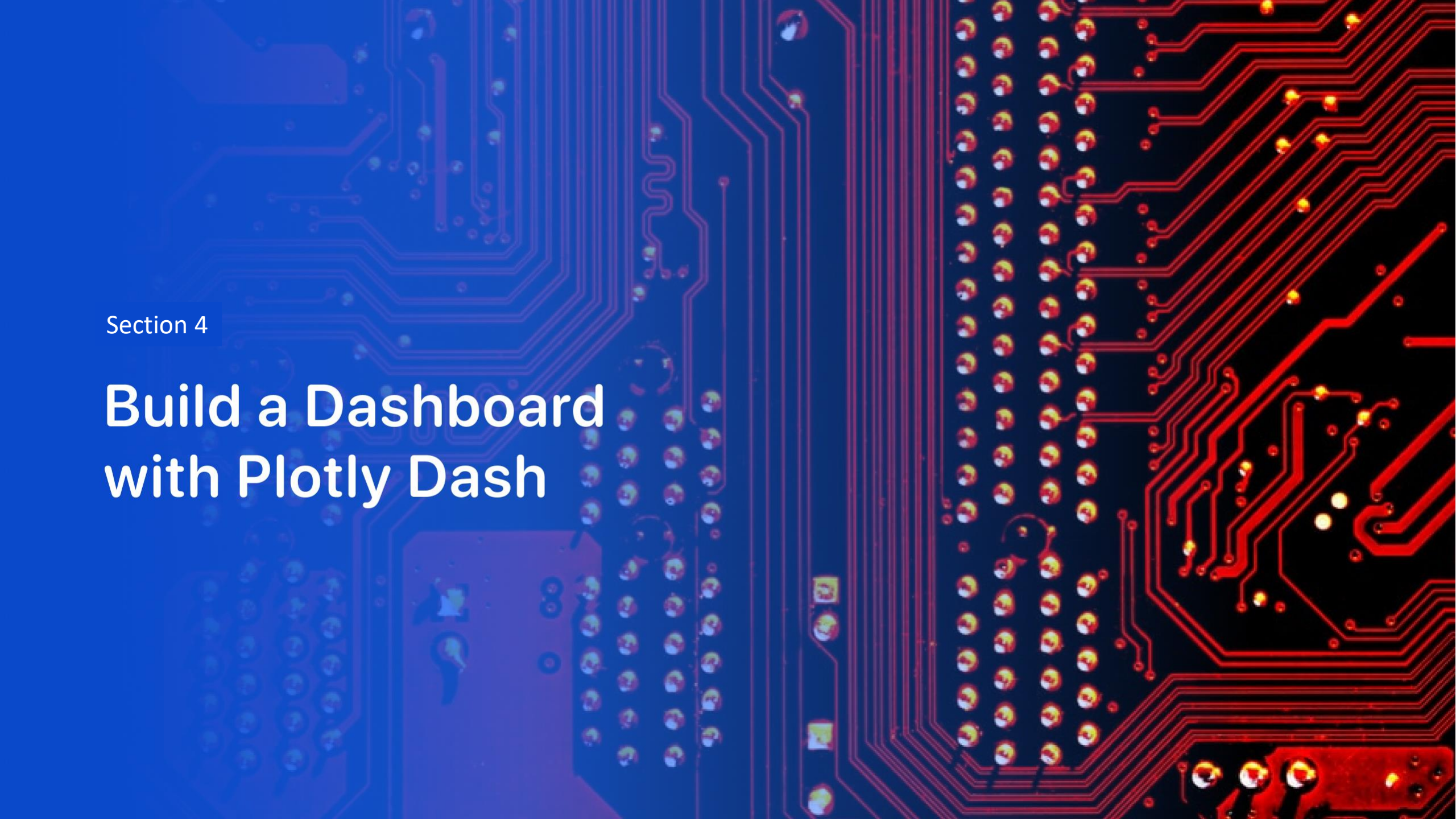


Green = Success, Red = Failure.

West Coast Launch Site Proximity Analysis



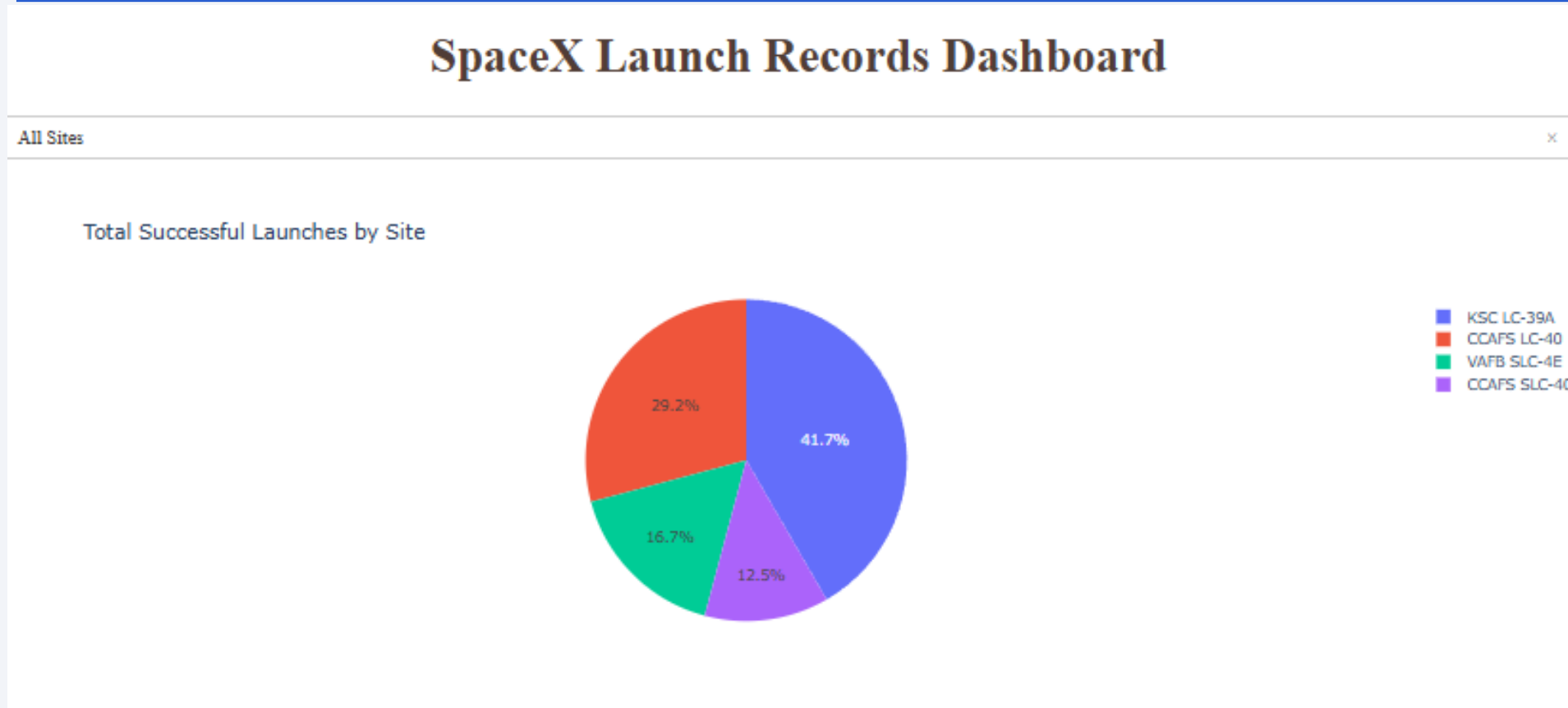
The launch site is close to the coastline and railway, but farther from the nearest city and highway.



Section 4

Build a Dashboard with Plotly Dash

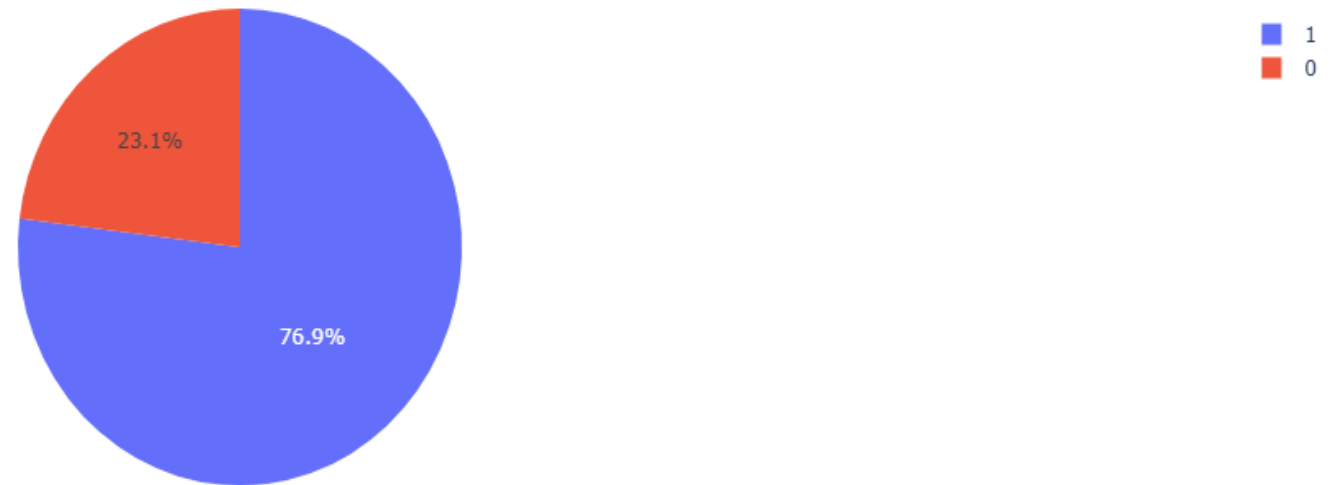
Successful Launches by Site (%)



The pie chart shows the percentage distribution of successful launches by launch site. KSC LC-39A leads with almost 42% of all successful launches.

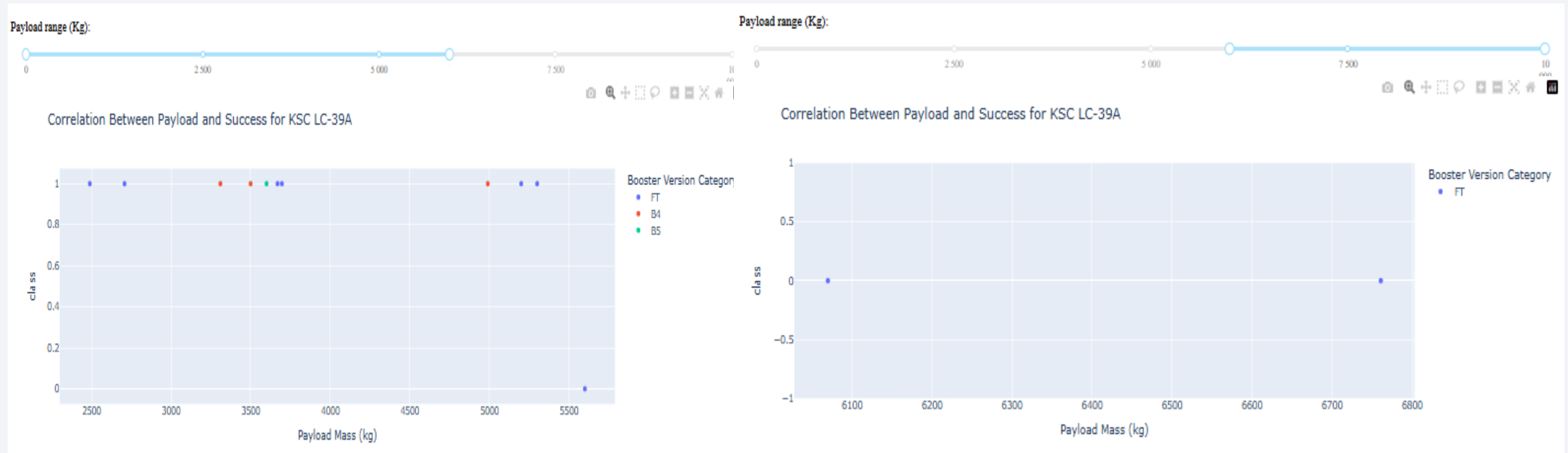
KSC LC-39A Launch site

Success vs Failed Launches for KSC LC-39A



KSC LC-39A has a launch success rate of 76.9%, making it one of the strongest-performing launch sites.

Payload Mass vs. Launch Outcome for KSC LC-39A



For KSC LC-39A, payloads between about 2,500 and 5,400 kg had the highest success rate, while payloads above 5,500 kg showed more failures. Booster versions B4 and B5 showed strong success performance in this selected view.



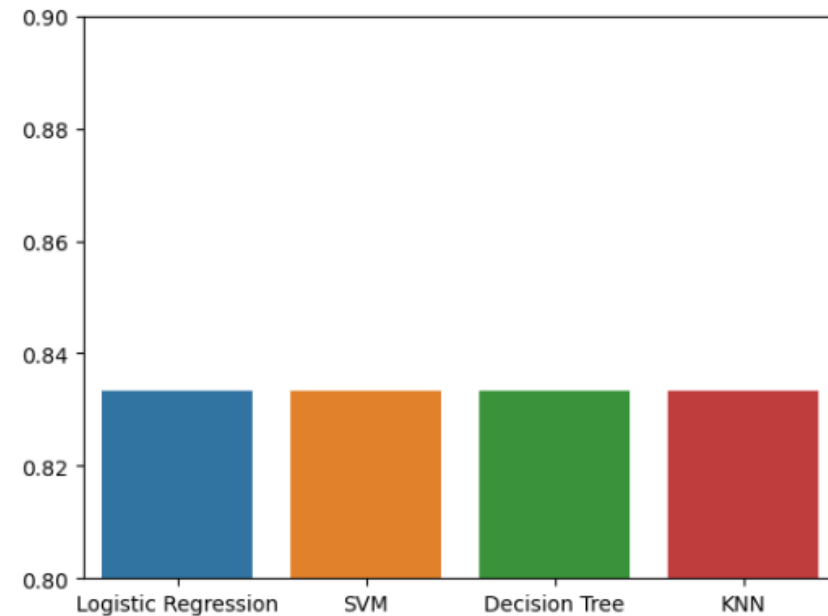
Section 5

Predictive Analysis (Classification)

Classification Accuracy

```
[95]: scores = {  
    'Logistic Regression': logreg_cv.score(X_test, Y_test),  
    'SVM': svm_cv.score(X_test, Y_test),  
    'Decision Tree': tree_cv.score(X_test, Y_test),  
    'KNN': KNN_cv.score(X_test, Y_test)  
}  
  
for model, score in scores.items():  
    print(f"{model}: {score:.2f}")  
  
best_model = max(scores, key=scores.get)  
  
print("Best model:", best_model)  
print("Best accuracy:", scores[best_model])  
  
Logistic Regression: 0.83  
SVM: 0.83  
Decision Tree: 0.83  
KNN: 0.83  
Best model: Logistic Regression  
Best accuracy: 0.8333333333333334
```

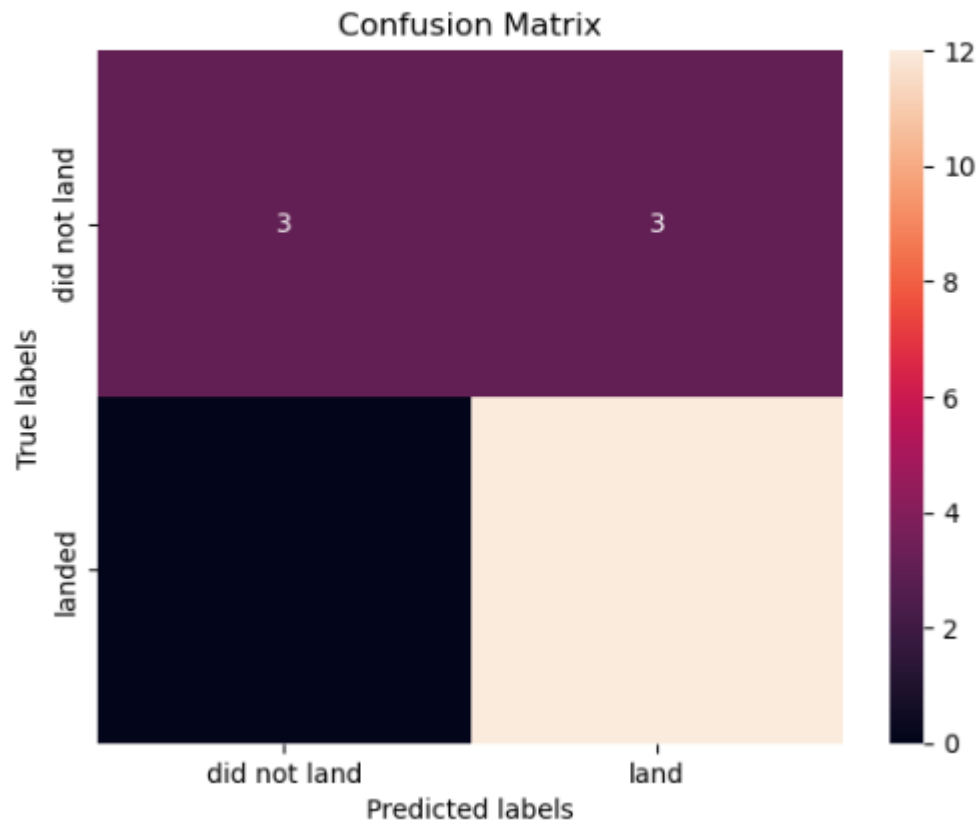
```
[81]: sns.barplot(x=list(scores.keys()), y=list(scores.values()))  
plt.ylim(0.8, 0.9)  
plt.show()
```



All four classification models—Logistic Regression, SVM, Decision Tree, and KNN—achieved the same accuracy of 83%.

Confusion Matrix

```
[74]: yhat=tree_cv.predict(X_test)
      plot_confusion_matrix(Y_test,yhat)
```



the Decision Tree model obtained the highest cross-validation score (`tree_cv.best_score_`). Therefore, the Decision Tree model was selected for further evaluation using the confusion matrix.

The confusion matrix shows that the model correctly classified 12 successful landings and 3 unsuccessful landings. However, 3 successful landings were incorrectly predicted as unsuccessful landings. No unsuccessful landings were incorrectly classified as successful. Overall, the model demonstrated good predictive performance, correctly classifying 83% of the test observations and effectively identifying both successful and failed Falcon 9 first-stage landings.

Conclusions

Based on the analysis, the following conclusions can be drawn:

- Launch sites with a higher number of flights generally exhibited higher landing success rates.
- The launch success rate showed an overall upward trend from 2013 to 2020, indicating improvements in Falcon 9 landing performance over time.
- The ES-L1, GEO, HEO, SSO, and VLEO orbit types achieved the highest landing success rates.
- KSC LC-39A recorded the greatest number of successful launches among all launch sites.
- Among the machine learning models evaluated, the Decision Tree classifier achieved the highest cross-validation score and was therefore identified as the most suitable algorithm for predicting Falcon 9 first-stage landing outcomes.

Appendix

Thank you!

